

# Danny Quang

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Github: [github.com/dquangucsd](https://github.com/dquangucsd)

Portfolio: [dquangucsd.github.io/dq-portfolio](https://dquangucsd.github.io/dq-portfolio)

## EDUCATION

- 
- **University of California, San Diego** San Diego, California  
• *MS - Computer Science, BS - Computer Science, GPA: 3.53* *Expected Graduation: 2026*

## WORK EXPERIENCE

- 
- **Intern for Software Engineering DevOps at Synaptics Inc.:** June 2025-September 2025   C, Python
    - Automated the audio test pipeline and designed the audio test harness. Wrote CI scripts for building, testing and reporting of wireless audio products. Saves about **720 engineering hours** in a year, or 2 hours a day **per engineer**
    - Prototyped channel sounding apparatus to implement remote long-distance automated testing of Bluetooth products. Saves about **540 engineering hours** a year, or about 1-2 hours a day, **per engineer**
    - Maintain Continuous Integration (CI) computer systems and test fixture for Bluetooth Low Energy firmware products
  - **Intern for Software Engineering at Wind River Systems, Inc.:** June 2023-September 2023   C, Python, Embedded C
    - Ported company's proprietary Real-Time Operating system (VxWorks7) to the Raspberry Pi Zero 2W
    - Designed a pinmux driver from scratch for the GPIO for general purpose ability and various other parts
    - VxWorks7 on the Pi Zero 2W - a commercial low power system - enables customers to run low cost, low power tasks
  - **Tutor for UCSD CSE: CSE 110, CSE 29:** March 2023-June 2023, March 2023-June 2024   Javascript, CSS, HTML, C
    - Tutor for CSE 110, Software Engineering: focuses on building a project with front end technologies (HTML, CSS, JS)
    - Tutor for CSE 29, Intro to Computer Systems: Lead Logistics Team. Focus on systems programming in C, software tools.
  - **Intern Data Analyst at The Center for Community Energy Inc.:** June-October 2022   Python, BeautifulSoup4, HTML
    - Web scraped the market involving V2G data, including multiple car company websites targeting key words
  - **Club Webmaster for American Institute of Aeronautics and Astronautics:** May 2022-November 2023   HTML, CSS
    - Redesigned, maintained, and updated the website for elected positions and events for AIAA UCSD

## PROJECTS

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- **Privacy Airtag:** C
    - Developed firmware for STM32 (ARM Cortex M4) board to act like an Airtag. Use Bluetooth Low Energy (BLE) driver to periodically broadcast signals to receiver when stationary for over 1 minute, turn off when moving to protect privacy
    - Detected movement with LSM6DSL accelerometer, wake when there is significant, successive movement
    - Minimized power consumption by **40%** by turning off all components when possible except for the accelerometer, and only waking necessary components on significant interrupt events
  - **Quadcopter:** C, C++, Arduino, Fusion360
    - Designed schematic and laid out board in Fusion 360, and developed firmware and proportional-integral-derivative (PID) tuning for flight with Arduino
    - Designed firmware to reliably send and receive signals for control (using PID controller) and feedback (filtered inertial measurement unit (IMU) signal)
  - **Twitter Clone:** Rust
    - Developed a distributed, load-balanced, scalable, fault-tolerant Twitter clone
    - Connected front end requests to the backend servers with clients using consistent hashing, formatted RPC calls
    - Implemented a keeper to for logical time and sync between the servers (inspired by Frangipani's Petal)
    - Used the async trait to meet performance requirements, like signing up 100 users concurrently within 5 seconds
  - **UGV:** Python, C++, ROS2
    - Designed firmware for an unmanned ground vehicle. First, implemented controls for speed and turning, with tuning for input to speed
    - Used camera to take in visual input to map to objects and adjust the vehicle's input and path trajectory

## SKILLS

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- **Languages:**     Python, Java, C, C++, Assembly, R, Swift, HTML, JavaScript, CSS, Node.js, React, SQL, Golang, Rust, Haskell, React.js, MongoDB, ROS
  - **Frameworks:**   ScikitLearn, Scipy, Matplotlib, Numpy, Pandas, Tensorflow (Keras), RegEx
  - **Tools:**         LaTeX, Vim, GitHub, Jupyter Notebook, VSCode, UNIX, Git, JUnit, AWS EC2, Xilinx, Vivado

***Relevant Coursework:** Graduate Computer Networks, Computer Security, Computer Vision, Databases, Operating Systems, Compilers, Computer Architecture, Advanced Data Structures and Algorithms, Software Engineering, Multivariate Statistical Learning, Representation Learning, Programming Languages, Web Client Languages, Robotic System Design and Implementation, Distributed Systems, Differentiable Programming, Parallel Programming, Parallel Programming for FPGA*